Group Members

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1. Project Summary

Habit Mastery League is a mobile habit-tracking application designed to help students build and maintain productive routines. Many students struggle to stay consistent with habits such as studying, exercising, or maintaining healthy routines. This application provides a structured and motivational way for users to track daily habits, monitor streaks, and analyze their progress over time.

The app allows users to create, edit, and manage habits through a simple and intuitive interface. Users can mark habits as completed from the dashboard, view detailed progress information for each habit, and analyze performance trends through statistics and insights screens. Motivational features such as streak tracking, milestone badges, completion heatmaps, and rule-based habit insights help encourage long-term consistency.

All data is stored locally on the device to ensure the application functions fully offline. Habit information and completion history are stored using a **SQLite database**, allowing the app to perform CRUD operations such as creating habits, updating progress, and deleting records. User preferences such as theme settings, reminder notifications, and motivational features are stored using **SharedPreferences**, ensuring that user settings persist across application sessions.

The expected outcome of the project is a fully functional mobile application that allows users to manage personal habits efficiently while demonstrating proper mobile development practices, including responsive UI design, local data persistence, and structured navigation.

3. Github Repository Link

Link: https://github.com/Myriad21/Team_U28_Habit_Mastery_League

5. Proposed Wireframe

Layout Screen Map:

Splash Screen

↓
Dashboard (Today's Habits)
↓
Add Habit Screen
↓
Habit Detail Screen
↓
Statistics / Insights Screen
↓
Settings Screen

Navigation Flow

- The **Splash Screen** loads the application and initializes local preferences.
- After loading, users are directed to the **Dashboard**, which displays today's habits and streak progress.
- Users can create new habits by clicking the add habit button which sends you to the **Add Habit Screen**.
- Selecting or editing a habit opens the **Habit Detail Screen**, where progress history and milestones are displayed.
- The **Statistics Screen** provides weekly insights and completion trends across all habits.
- The **Settings Screen** allows users to modify preferences such as theme settings and user profile information.

5.1 - Splash Screen

Purpose:

The Splash Screen is displayed when the application first launches. It briefly presents the app branding and initializes local data before navigating to the main dashboard.

Navigation Flow:
Splash Screen → Dashboard

Key UI Components:

- App logo and Title
- Loading indicator
- Background container

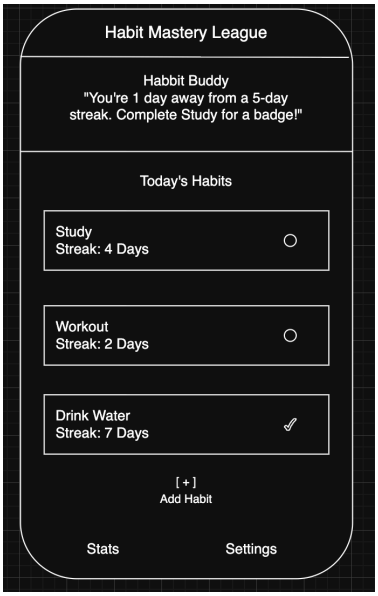
User Interaction:
This screen does not require user interaction. After initialization completes, the application automatically navigates to the Dashboard screen.

5.2 - Dashboard

Purpose

The Dashboard serves as the central hub of the Habit Mastery League application. It allows users to quickly view their daily habits, track current streaks, and mark habits as completed. The dashboard prioritizes simplicity and speed so users can interact with their habits in a single tap.

This screen also provides navigation to other areas of the application including creating, editing, and deleting habits as well as statistics, and application settings.



Navigation Flow:

Splash Screen → Dashboard,

Dashboard → Add Habit Screen

Dashboard → Habit Detail Screen

Dashboard → Statistics Screen

Dashboard → Settings Screen

Key UI Components:

- App Header: Displays the application title *Habit Mastery League*
- Habit Buddy Message**:** A rule-based motivational message that provides encouragement or reminders based on the user's recent progress. Eg. "You're one day away from a 5-day streak. Complete Study to unlock a badge."
- Today's Habits List: A scrollable list of habit cards displaying:
    - Habit name
    - Current streak count
    - Completion status

  Each habit card visually indicates whether the habit has been completed for the day.

- Add Habit Button: A button that opens the Add Habit screen where users can create new habits.
- Bottom Navigation: Quick navigation links to:
    - Statistics Screen
    - Settings Screen

User Interaction

Users interact with habit cards using gesture-based controls.

Tap the Habit card: Marks the habit as completed for the current day and updates the streak count.

Swipe Left on Habit Card: Reveals action options

- View Habit Details → Habit Details Screen
- Edit Habit → Add Habit Screen
- Delete Habit

Add Habit Button: Opens the Add Habit screen

Bottom Navigation: Allows users to switch between Dashboard, Statistics, and Settings screens.

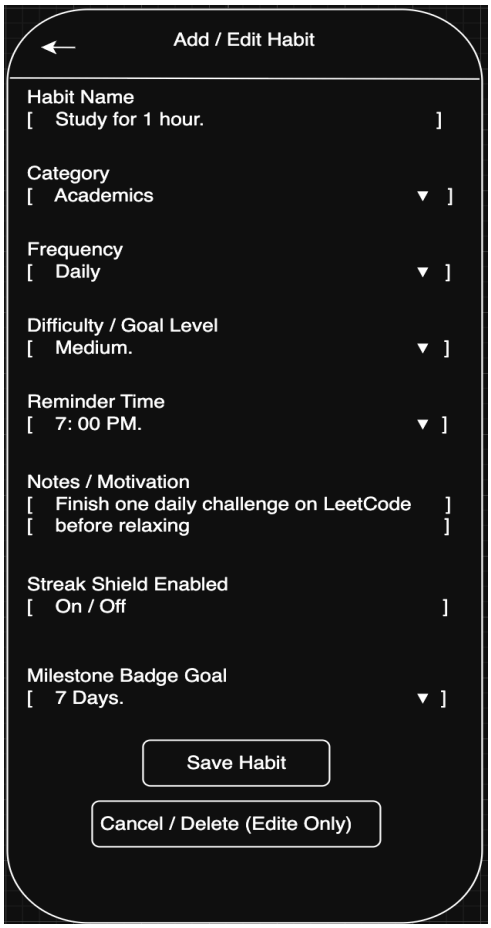
Data Flow

The Dashboard retrieves and loads the following data from the SQLite database: Habit List, Habit completion status for the current day, and Streak counts. Completion actions update the habit_logs table, which records daily habit completion.

Edge Cases

1. Empty Habit list: If the user has not created any habits, the dashboard will display the message: "No habits created yet. Tap + to add your first habit."
2. Completion Update Error: If the application fails to record a habit completion due to a database error, the user will receive a notification prompting them to try again.

5.3 - Add / Edit Habit Screen



Purpose

The Add / Edit Habit screen allows users to create new habits or modify existing ones. This screen provides a structured form where users can define the behavior they want to track, set reminders, and configure motivational features such as milestone badges and streak protection.

The same interface is used for both creating and editing habits. When accessed through the Add Habit button, the form fields are empty. When accessed through the Edit option on a habit card, the form is pre-filled with the existing habit data.

Navigation Flow:

Dashboard → Add Habit Screen

Dashboard → Edit Habit → Add/Edit Habit Screen

Add/Edit Habit Screen → Save → Dashboard

Add/Edit Habit Screen → Cancel → Dashboard

Key UI Components:

- Habit Name: A text input field where the user enters the name of the habit they want to track.

- Category (Dropdown Menu): Allows users to categorize their habits for better organization. Options include:
    - Academics, Fitness, Health, Productivity, Personal Development, Other
- Frequency (Dropdown Menu): Defines how often the habit should be completed. Options include:
    - Daily, Weekdays, Weekends, Custom Schedule
- Difficulty / Goal Level (Dropdown Menu): Allows users to define the effort level associated with the habit. Options include:
    - Easy, medium, Hard
- Reminder Time (Dropdown Menu): Allows users to set a daily reminder for completing the habit. OPtions include:
    - All times
- Notes / Motivation: A multi-line text field that allows users to write a short motivational message or reminder.
- Streak Shield Toggle: A toggle switch that enables a streak protection feature, When enabled, users may skip a habit once without losing their streak.
- Milestone Badge Goal (Dropdown Menu). OPtions Include:
    - 3 days, 7 Days, 14 Days, 30 Days, custom.
- Save Habit Button: Stores the habit configuration in the SQLite database.
- Cancel / Delete Button: Depending on context provides two different actions.
    - Cancel: Returns the user to the dashboard without saving changes.
    - Delete (Edit Mode Only): Removes the habit from the database

User Interaction

Users complete the form fields and press Save Habit to create or update a habit.

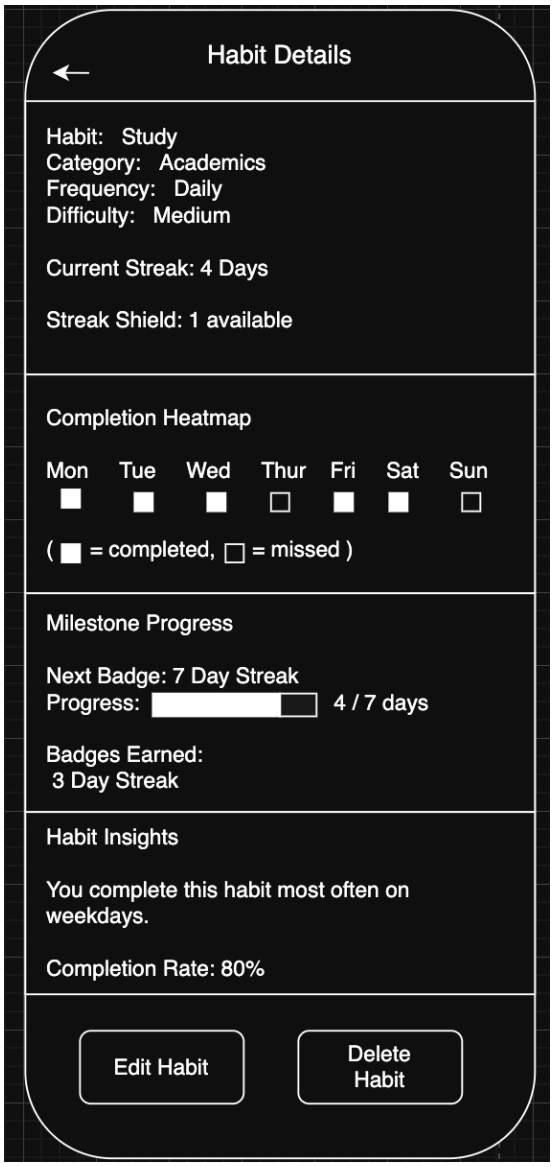
Data Flow

This screen interacts directly with the SQLite database, when a habit is saved: Habit details are stored in the habits table, Habit configuration data is saved for future tracking and analytics.

Edge Cases

1. Empty Habit Name: If the habit name field is empty, the application will display a validation message: “Habit name cannot be empty.” The habit will not be saved until a valid name is entered.
2. Failed Save Operation: If the database fails to save the habit due to an unexpected error, the user will receive a message prompting them to try again.

5.4 - Habit Detail Screen



Purpose

The Habit Detail screen provides users with detailed information about a specific habit. It allows users to monitor their progress, view historical completion data, track streaks, and review achievements earned through consistent habit completion.

This screen helps users understand their performance patterns and encourages long-term habit development through visual feedback and milestone tracking.

Navigation Flow:

Dashboard → View Habit → Habit Details Screen

Edit Habit → Add / Edit Habit Screen

Delete Habit →

Key UI Components:

- Habit Information: The top section of the screen displays the core details of the habit which displays: Habit Name, Category, Frequency, and Difficulty
- Streak Tracking: The screen displays the user's current habit streak, which represents the number of consecutive days the habit has been completed.
- Streak Shield Status: The streak shield feature allows a user to miss a day without resetting their streak. The screen displays whether a streak shield is currently available.
- Completion Heatmap: The completion heatmap visually represents habit activity across a week. Filled boxes mean the habit was completed while empty boxes

mean habit was missed. The heatmap data is retrieved from the habit_logs table in the SQLite database.

- Milestone Progress: This section shows progress toward the next milestone badge with a progress bar.
- Badges Earned: This section displays badges that have already been unlocked through previous milestones
- Habit Insights: The Habit Insights section provides a short analysis of the user's habit behavior.
- Edit Habit Button: The Edit Habit button allows users to modify the habit configuration by sending them to the Add/Edit Habit screen.
- Delete Habit Button: The Delete Habit button allows users to permanently remove a habit. A confirmation dialog should appear before deletion to prevent accidental removal.

User Interaction

Users primarily interact with this screen by reviewing progress and managing their habits. Some common actions include: Viewing completion history, Tracking streak progress, Reviewing milestone achievements, Editing habit configuration, and Deleting habits if they are no longer needed.

Data Flow

This screen retrieves data from certain tables in the SQLite database. Such as:

- Habits: Stores habit configuration data such as name, category, frequency, and difficulty.
- Habit_logs: Stores daily completion records used to calculate streaks and populate the completion heatmap. This data is processed to generate streak values, completion rates, and progress toward milestone badges.

Edge Cases

1. No Completion Data: If a habit has been created but has not yet been completed, the heatmap will display empty entries. Eg. “No completion data available yet. Start completing this habit to track your progress.“
2. No Badges Earned: If the user has not yet reached any milestones, the badge section will display a placeholder message: “No badges earned yet. Complete your habit consistently to unlock achievements.”

5.5 - Statistics Screen

Purpose

The Statistics / Insights screen provides users with a comprehensive overview of their habit performance across all tracked habits. This screen aggregates habit data and presents it in a clear format that helps users understand patterns, track improvement, and stay motivated.

By visualizing completion rates, streak rankings, and weekly trends, this screen helps users evaluate their consistency and identify areas for improvement.

Navigation Flow:

Dashboard → Statistics Screen

Statistics Screen → Dashboard

Key UI Components:

- Weekly Habit Score: This section displays the completion percentage for each habit during the current week. Progress bars visually represent each habit's completion rate. These values are calculated using completion records stored in the database.
- Weekly Completion Trend: This section displays the number of habits completed each day of the week.
- Streak Leaderboard: The Streak Leaderboard ranks habits based on their current streak length.
- Habit Insights: The Habit Insights section provides a short analysis of the user's behavior patterns. These insights are generated using rule-based analysis of habit completion data stored in the database.

- Weekly Reflection: This section provides a motivational summary comparing the user's current performance with previous activity.

User Interaction

Users primarily interact with this screen by reviewing statistics and progress summaries. Common actions include: Monitoring weekly completion trends, Comparing habit performance, Viewing streak rankings, Reviewing personalized insights.

This screen is designed primarily for information display rather than direct habit interaction.

Data Flow

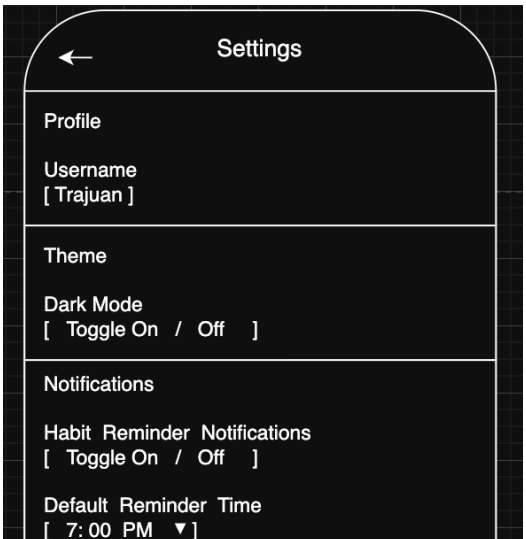
This screen retrieves specific data from certain tables in the SQLite database. Such as:

- habits: Stores habit definitions such as habit name and configuration
- habit_logs: Stores daily completion records used to calculate: completion percentages, streak rankings, weekly activity trends, average completion rate.

Edge Cases

1. No Habit Data: If no habits have been created, the screen will display a message indicating that no statistics are available. Eg. “No habit data available yet. Create habits and start tracking progress to view statistics.”
2. No Completion Data: If habits exist but no completions have been recorded yet, the statistics will show placeholder values until activity data becomes available. Eg. “No completion data recorded yet. Complete your habits to generate insights.”

5.6 - Settings Screen



Purpose

The Settings screen allows users to personalize their experience within the Habit Mastery League application. This screen provides controls for profile information, visual preferences, notification settings, and data management.

User preferences configured on this screen are stored locally using SharedPreferences, allowing settings to persist between application sessions.

Navigation Flow:

Dashboard → Settings Screen

Settings Screen → Dashboard

Key UI Components:

- Profile: This section allows the user to set a personal username. The username is stored locally and may be used to personalize the application interface or motivational messages.
- Theme Settings: This section controls the visual appearance of the application.
    - Dark Mode Toggle**:** Users can switch between light and dark themes. The selected theme preference is stored locally and applied whenever the application is launched.
- Notification Settings: This section controls reminder notifications for habit completion.
    - Habit Reminder Notifications: Allows users to enable or disable reminder notifications.
    - Default Reminder Time: Allows users to select a default reminder time that will automatically be applied when new habits are created. Eg. 7PM, 8PM etc.
- Habit Preferences: This section allows users to enable or disable motivational features within the application.
    - Habit Buddy Suggestions: Controls whether the application generates motivational suggestions based on habit progress.
    - Weekly Reflection Messages: Allows users to enable or disable weekly performance summaries.

- Data Management: This section provides tools for managing stored habit data.
    - Reset All Habit Data: Allows the user to remove all stored habits and completion records. When selected, the application should display a confirmation dialog before proceeding to prevent accidental deletion.
- App Information: Displays basic application information such as version and title.

User Interaction

Users interact with this screen primarily by adjusting toggles and preference values. Typical actions include: Changing the application theme, Enabling or disabling notifications, Configuring reminder times, Enabling motivational features, and Resetting habit data if needed.

Data Storage

User preferences on this screen are stored using SharedPreferences.

Eg. username, darkModeEnabled, notificationsEnabled, etc

These values are loaded when the application starts and applied to the interface accordingly.

Edge Cases

1. Reset Confirmation: If a user attempts to reset all habit data, the application should display a confirmation prompt. Eg. “Are you sure you want to delete all habit data? This action cannot be undone.”
2. Disabled Notifications: If notifications are disabled, reminder alerts should not be triggered even if habits have reminder times configured.

6. Iterative Design and Testing Plan

To ensure the Habit Mastery League application is intuitive, functional, and easy to use, the design will go through an iterative testing process. Initial wireframes were created using low-fidelity layouts that focus on structure and functionality rather than visual styling. These wireframes allow the development team to evaluate navigation flow, usability, and screen organization before implementing the final interface in Flutter.

User feedback and testing will be used to refine the interface and improve the overall experience. The goal of this process is to identify usability issues early and adjust the design before development is finalized.

Test Case 1: Habit Creation Flow

Users will attempt to create a new habit using the Add/Edit Habit screen.

Expected Result:
Users should be able to enter habit information, select options from dropdown menus, and successfully save the habit without confusion.

Revision Plan:
If users struggle to understand the form fields or dropdown menus, the layout will be simplified, labels will be clarified, and spacing between fields will be improved.

Test Case 2: Habit Completion Interaction

Users will mark habits as completed from the Dashboard screen.

Expected Result:
Users should be able to quickly mark habits complete using a single tap, and the interface should clearly update the completion status and streak count.

Revision Plan:
If users have difficulty understanding how to complete a habit, visual indicators such as clearer completion icons or animation feedback may be added.

Test Case 3: Navigation Between Screens

Users will navigate between the Dashboard, Statistics, Habit Details, and Settings screens.

Expected Result:
Navigation should be straightforward, and users should easily understand how to move between screens using buttons or navigation controls.

Revision Plan:
If users have difficulty locating navigation options, button placement and navigation labels will be adjusted to improve clarity.

Test Case 4: Statistics and Progress Understanding

Users will view the Statistics screen to interpret their habit progress.

Expected Result:
Users should easily understand completion percentages, weekly trends, and streak rankings.

Revision Plan:
If users find the statistics confusing, visual elements such as clearer labels, improved spacing, or simplified charts will be implemented.

Edge Case Testing

Additional tests will ensure the application behaves properly in uncommon situations.

Examples include:

• No habits created — the Dashboard and Statistics screens should display clear instructions encouraging the user to create their first habit.
• Invalid habit input — the application should prevent saving a habit without a name and display an error message.
• Failed database save — the application should notify the user and prompt them to retry the action.